جامعة الشرقية
A' SHARQIYAH UNIVERSITY

INTE130 – Object-Oriented Programming Assignment.

(CyberShield system Project)

Sumitted Parts: Part C

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Topics of the project:

1- System structure.

2- Multi Agent Structure.

3- System Classes.

4- Explaining the relationships between the Classes.

5- UML Class Diagram.

6- UML diagram explanation.

7- Conclusion.

1- System structure:

- **CyberShield's system is based on the concept of a one-person company assisted by a team of digital agents. In this system, the founder is primarily responsible for inputting tasks, reviewing results, and making final decisions. Each digital agent then handles a specific part of the task.**

- **The system process begins when the founder creates a task, such as a system security check. This task is then sent to digital agents, each of whom analyzes the task according to their role and presents the analysis results or appropriate recommendations. Finally, the founder reviews all the results and makes the final decision based on them.**

- **This approach helps to organize work and divide responsibilities within the system, and makes managing the company easier and more efficient.**

2- Multi Agent Structure:

❖ **The system consists of three main agents, each with a specific role:**

1- Threat monitoring agent:

Responsibility: Monitoring the system and detecting suspicious activities.

Type of task: Checking for abnormal activity or unusual login attempts.

Processed information: Activity data within the system.

2- Vulnerability Analysis Agent:

Responsibility: To discover weaknesses in the system.

Type of task: Checking weak passwords or settings.

Information processed: Protection information and system status.

3- Response Agent:

Responsibility: To propose appropriate solutions when a problem is discovered.

Type of task: Performing actions such as changing passwords or blocking access.

Information processed: Results of other agents and level of problem.

3- System Classes:

- ❖ **The system consists of several main classes, which are:**

1- Founder :

The person in charge of the system is responsible for creating the task, reviewing the results, and making the final decision.

2- Task :

This represents the work that needs to be performed, such as checking the system's security.

3- ThreatAgent :

The agent is responsible for detecting threats and suspicious activities.

4- VulnerabilityAgent:

The agent is responsible for analyzing gaps and weaknesses.

5- ResponseAgent:

The agent is responsible for proposing appropriate solutions.

4- Explaining the relationships between the Classes:

- **The relationship between the system components is as follows:**

1- **The founder creates the task.**

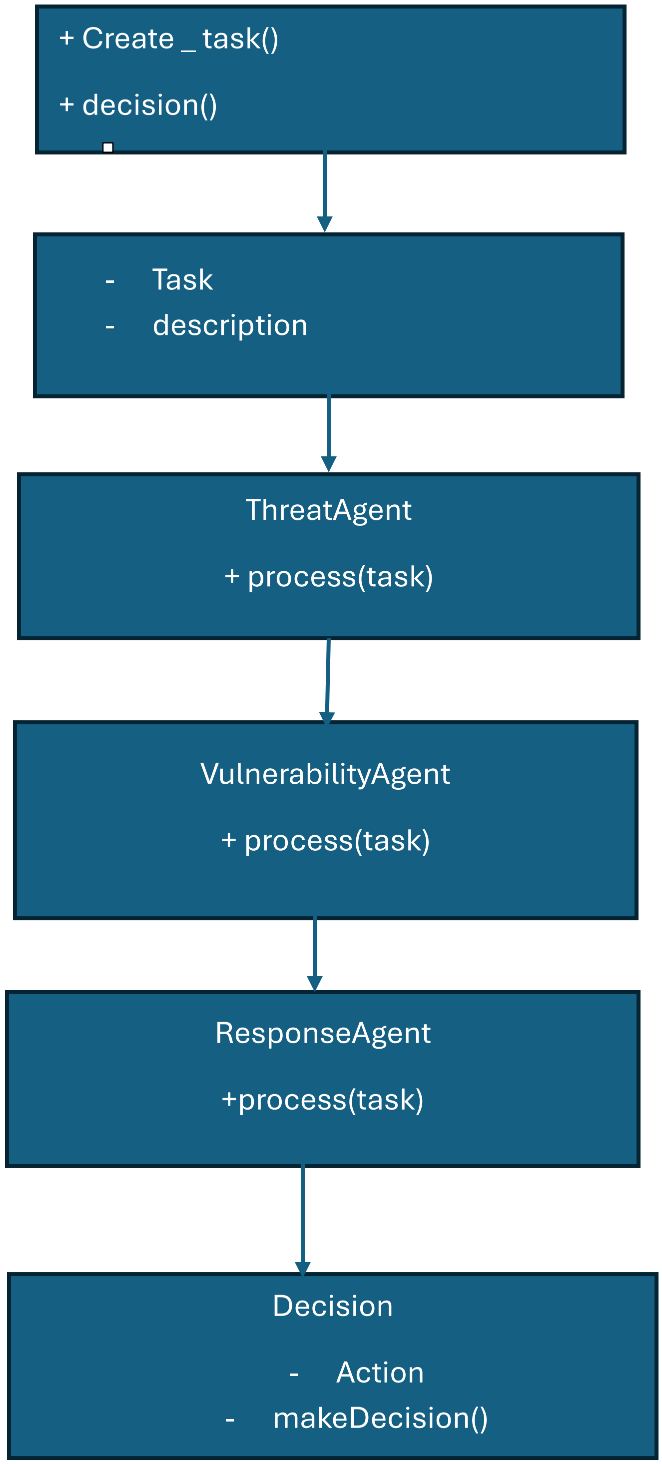
2- **The task is sent to the agents.**

3- **Each agent analyzes the task according to their job.**

4- **The agents are displaying the results.**

5- **The founder reviews the results and makes the final decision.**

5 - UML Class Diagram :



6- UML diagram explanation:

The UML diagram shows that the founder is responsible for creating the task. The task is then handled by three digital agents: a threat monitoring agent, a vulnerability analysis agent, and a response agent. Each agent has a process(task) function to process the task according to its function. Finally, the founder uses the results to make the final decision.

7- Conclusion:

The CyberShield system design demonstrates how work can be divided within a single-person company using multiple digital agents. This section also shows that the system is organized and clear, with each component having a defined role, facilitating the transition to the code implementation phase in the next section.